

Mathematical Correctness of Sufficient-Statistics OLS

sufficient-regression design note

June 30, 2026

1 Scope and assumptions

This note gives the algebra that an IncrementalOLS, RollingOLS, weighted OLS, ridge, and exponentially forgotten OLS implementation can share. The design matrix is

$$X = \begin{bmatrix} x_1^T \\ \vdots \\ x_n^T \end{bmatrix} \in \mathbb{R}^{n \times p}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n.$$

Each row vector $x_i \in \mathbb{R}^p$ must already contain any intercept column. The formulas do not add an intercept implicitly.

The sufficient statistics are

$$G = X^T X = \sum_{i=1}^n x_i x_i^T, \quad h = X^T y = \sum_{i=1}^n x_i y_i, \quad q = y^T y = \sum_{i=1}^n y_i^2, \quad n.$$

The matrix G is symmetric positive semidefinite. Ordinary least squares has a unique coefficient vector exactly when G is positive definite, equivalently when X has full column rank. In formulas below, inverses denote mathematical solutions. An implementation should solve linear systems using a numerically appropriate factorization and should fail loudly when the stated rank or definiteness conditions are not met.

2 Ordinary least squares from sufficient statistics

Lemma 1 (Objective expansion). *For any coefficient vector $\beta \in \mathbb{R}^p$,*

$$\|y - X\beta\|_2^2 = q - 2\beta^T h + \beta^T G\beta.$$

Proof. Expand the quadratic:

$$(y - X\beta)^T (y - X\beta) = y^T y - 2\beta^T X^T y + \beta^T X^T X\beta = q - 2\beta^T h + \beta^T G\beta.$$

□

Theorem 1 (OLS coefficient formula). *If G is positive definite, the unique ordinary least squares solution is*

$$\hat{\beta} = G^{-1}h.$$

Proof. By the objective expansion, the gradient is

$$\nabla_{\beta} (q - 2\beta^T h + \beta^T G \beta) = -2h + 2G\beta.$$

Setting the gradient to zero gives the normal equation

$$G\hat{\beta} = h.$$

Positive definiteness makes the objective strictly convex, so this stationary point is the unique minimizer. $\square$

Corollary 1 (Residual sum of squares). *For any candidate β ,*

$$\text{RSS}(\beta) = q - 2\beta^T h + \beta^T G \beta.$$

For the OLS solution,

$$\text{RSS}(\hat{\beta}) = q - h^T G^{-1} h.$$

Proof. The first identity is the objective expansion. For the minimizer, $G\hat{\beta} = h$, so

$$\hat{\beta}^T G \hat{\beta} = \hat{\beta}^T h = h^T G^{-1} h.$$

Substitution gives the result. $\square$

Therefore two data sets with the same (G, h, q, n) have the same OLS objective for every β , the same OLS coefficients when the solution is unique, and the same residual sum of squares. Raw rows are not required for these quantities once the statistics are correct.

3 IncrementalOLS append formulas

Let $S_t = (G_t, h_t, q_t, n_t)$ be the statistics after t rows. When a new observation (x, y) arrives,

$$G_{t+1} = G_t + xx^T, \quad h_{t+1} = h_t + xy, \quad q_{t+1} = q_t + y^2, \quad n_{t+1} = n_t + 1.$$

For a batch $X_B \in \mathbb{R}^{m \times p}$, $y_B \in \mathbb{R}^m$,

$$G_{\text{new}} = G_{\text{old}} + X_B^T X_B, \quad h_{\text{new}} = h_{\text{old}} + X_B^T y_B, \quad q_{\text{new}} = q_{\text{old}} + y_B^T y_B, \quad n_{\text{new}} = n_{\text{old}} + m.$$

Theorem 2 (Correctness of append). *The append formulas produce exactly the same sufficient statistics as recomputing $X^T X$, $X^T y$, and $y^T y$ from all rows after the append.*

Proof. The statistic G is a sum of row outer products. Appending (x, y) changes that sum by exactly xx^T :

$$\sum_{i=1}^{t+1} x_i x_i^T = \sum_{i=1}^t x_i x_i^T + x_{t+1} x_{t+1}^T.$$

The same additive argument applies to $h = \sum_i x_i y_i$, $q = \sum_i y_i^2$, and n . The batch case follows by applying the single-row argument to each row in the batch, or directly from block matrix multiplication. $\square$

4 RollingOLS add/drop formulas

For a rolling window, the active data set is a finite multiset of rows. Suppose the current window statistics include an old observation (x_{old}, y_{old}) . Dropping it gives

$$G^- = G - x_{old}x_{old}^T, \quad h^- = h - x_{old}y_{old}, \quad q^- = q - y_{old}^2, \quad n^- = n - 1.$$

Adding a new observation (x_{new}, y_{new}) after the drop gives

$$G^+ = G^- + x_{new}x_{new}^T, \quad h^+ = h^- + x_{new}y_{new}, \quad q^+ = q^- + y_{new}^2, \quad n^+ = n^- + 1.$$

For a fixed-size one-step slide these combine to

$$G^+ = G - x_{old}x_{old}^T + x_{new}x_{new}^T, \\ h^+ = h - x_{old}y_{old} + x_{new}y_{new}, \quad q^+ = q - y_{old}^2 + y_{new}^2, \quad n^+ = n.$$

Theorem 3 (Correctness of add/drop). *If the dropped observation is in the active window, the add/drop formulas produce exactly the sufficient statistics of the new active window.*

Proof. The active-window statistic G is the sum of xx^T over exactly the rows in the window. Removing one row subtracts that row's summand, and adding one row adds the new summand. The same summand-by-summand argument applies to h , q , and n . $\square$

Rolling deletion cannot be inferred from (G, h, q, n) alone: the implementation must know the departing row, or at least its stored contributions xx^T , xy , and y^2 . This is a mathematical requirement, not an implementation preference.

5 Weighted OLS

Let $W = \text{diag}(w_1, \dots, w_n)$, with $w_i \geq 0$. The weighted least squares objective is

$$L_w(\beta) = (y - X\beta)^T W (y - X\beta) = \sum_{i=1}^n w_i (y_i - x_i^T \beta)^2.$$

Define weighted sufficient statistics

$$G_w = X^T W X = \sum_{i=1}^n w_i x_i x_i^T, \quad h_w = X^T W y = \sum_{i=1}^n w_i x_i y_i, \quad q_w = y^T W y = \sum_{i=1}^n w_i y_i^2.$$

If G_w is positive definite, then

$$\hat{\beta}_w = G_w^{-1} h_w.$$

Proof. The objective expands as

$$L_w(\beta) = q_w - 2\beta^T h_w + \beta^T G_w \beta.$$

The gradient is $-2h_w + 2G_w\beta$. Positive definiteness gives a strictly convex objective, so $G_w \hat{\beta}_w = h_w$ has the unique minimizer. $\square$

Equivalently, weighted OLS is unweighted OLS on transformed rows $\sqrt{w_i}x_i$ and transformed responses $\sqrt{w_i}y_i$. An incremental weighted append with observation weight w is

$$G_{new} = G + wx x^T, \quad h_{new} = h + wxy, \quad q_{new} = q + wy^2.$$

Weighted rolling add/drop uses the same signed contributions. Negative weights do not define a least-squares norm and should be rejected.

6 Ridge regression

Let $R \in \mathbb{R}^{p \times p}$ be a symmetric positive semidefinite penalty matrix. Ridge solves

$$\hat{\beta}_R = \arg \min_{\beta} [\|y - X\beta\|_2^2 + \beta^T R \beta].$$

Using the same sufficient statistics,

$$\|y - X\beta\|_2^2 + \beta^T R \beta = q - 2\beta^T h + \beta^T (G + R) \beta.$$

If $G + R$ is positive definite, then

$$\hat{\beta}_R = (G + R)^{-1} h.$$

Proof. The gradient of the penalized objective is

$$-2h + 2(G + R)\beta.$$

The normal equation is $(G + R)\hat{\beta}_R = h$. Positive definiteness of $G + R$ makes the penalized objective strictly convex, so the solution is unique. $\square$

Scalar ridge with all coefficients penalized uses $R = \lambda I_p$, $\lambda \geq 0$. If the intercept should not be penalized, the design matrix should place the intercept in a known column and use a penalty matrix such as

$$R = \text{diag}(0, \lambda, \dots, \lambda).$$

This convention must be explicit because it changes the estimator.

7 Exponential forgetting

Let $\alpha \in (0, 1]$ be a forgetting factor. At time t , define exponentially weighted statistics

$$G_t = \sum_{i=1}^t \alpha^{t-i} x_i x_i^T, \quad h_t = \sum_{i=1}^t \alpha^{t-i} x_i y_i, \quad q_t = \sum_{i=1}^t \alpha^{t-i} y_i^2.$$

The recurrence for a new observation (x_t, y_t) is

$$G_t = \alpha G_{t-1} + x_t x_t^T, \quad h_t = \alpha h_{t-1} + x_t y_t, \quad q_t = \alpha q_{t-1} + y_t^2.$$

The effective weight mass follows

$$s_t = \alpha s_{t-1} + 1, \quad s_0 = 0,$$

but s_t is not an unweighted row count when $\alpha < 1$.

Theorem 4 (Correctness of exponential forgetting). *The recurrence above maintains the exponentially weighted statistics defined by the closed-form sums. If G_t is positive definite, the forgotten OLS coefficient vector is*

$$\hat{\beta}_t = G_t^{-1} h_t.$$

Proof. Assume the closed form holds at time $t - 1$. Then

$$\alpha G_{t-1} + x_t x_t^T = \alpha \sum_{i=1}^{t-1} \alpha^{t-1-i} x_i x_i^T + x_t x_t^T = \sum_{i=1}^t \alpha^{t-i} x_i x_i^T.$$

The same induction proves the formulas for h_t and q_t . The coefficient formula then follows from the weighted OLS theorem with weights α^{t-i} . $\square$

With both external weights and forgetting, a new weighted observation (x_t, y_t, w_t) contributes

$$G_t = \alpha G_{t-1} + w_t x_t x_t^T, \quad h_t = \alpha h_{t-1} + w_t x_t y_t, \quad q_t = \alpha q_{t-1} + w_t y_t^2.$$

8 Summary

Model or operation	Statistic update or solve
OLS	$\hat{\beta} = G^{-1}h$ when $G \succ 0$.
Append	Add xx^T , xy , y^2 , and one row count.
Rolling drop	Subtract the departing row's xx^T , xy , and y^2 .
Weighted OLS	Use $G_w = \sum_i w_i x_i x_i^T$, $h_w = \sum_i w_i x_i y_i$, $q_w = \sum_i w_i y_i^2$.
Ridge	Solve $(G + R)\hat{\beta} = h$.
Forgetting	Scale previous statistics by α , then append the new contribution.